

Review of Imre Lakatos's *Proofs and Refutations*

1977

Lakatos, Imre. . . . *Proofs and Refutations: The Logic of Mathematical Discovery*. (Edited by John Worrall and Elie Zahar.) London: Cambridge University Press, 1976. £7.50 cloth, £1.95 paperback. Pp. xii, 174.

This posthumous volume is a supplemented reissue of the similarly entitled essay that appeared in four parts in *The British Journal for the Philosophy of Science* in 1963–4.

The hundred pages of that remarkable essay ring changes on a single geometrical theme: Euler's law that the faces and vertices of a polyhedron together outnumber the edges by two. After explaining the classical proof, Lakatos produces an exception: a hollow solid whose surfaces are a cube within a cube. Its faces and vertices outnumber its edges by four. Then he examines the classical proof to see how it falls foul of such examples, and what stipulations would be suitable for excluding them. Having thus narrowed the scope of Euler's law, he produces a further exception: a solid consisting of two tetrahedra with only an edge or vertex in common. A further tightening of the law is thus indicated, and still the exceptions are not at an end. A polyhedron with a square tunnel through it occasions a further restriction; a cube with a penthouse on top occasions yet a further restriction; and so the dialectic of revision and exception goes its oscillating way.

The geometry is fascinating, but the purpose is philosophical. Lakatos is opposing the formalists' conception of mathematical proofs, which represents them as effectively testable and, once tested, incontrovertible. He is op-

posing the notion, so central to logical positivism, that mathematics and natural science are methodologically unlike.

In respect of sprightliness the style contrasts markedly with the subject matter. The text is a spirited dialogue among a teacher and sixteen pupils. The footnotes, nearly as voluminous as the text, furnish historical precedents for ideas and attitudes expressed in the dialogue. Lengthiness of footnotes is in most writings a sign of poor organisation: failure to worry one's material coherently into one-dimensional prose. But not so here, where the two levels of the page distinguish systematically between the fictitious participants of the dialogue and real mathematicians of the past three centuries. The lower level is fully as rewarding as the upper, the invidious distinction between nine- and twelve-point type notwithstanding. The wealth of scholarship is overwhelming.

The successive amputations from Euler's law are cunningly contrived and ordered, so as not to disqualify too many counter-instances at any one time. A major amputation early in the treatment would mean losing most of the illustrations of the methodology of theory construction, as well as spoiling the fun. At points this prolonging of curtailments is strained. Thus one counter-instance that Lakatos exploits is Kepler's star polyhedron, viewed as bounded by only twelve faces each of which is a star with an empty pentagonal centre. The more natural view of it, as bounded by sixty triangular faces, is allowed to emerge only later. The excuse for the first or dodecahedral account of the star polyhedron is that the five triangles composing each hollow star all lie in one plane, as befits a face. Yet in the fullness of time even the planarity of faces conveniently lapses, when a cylinder, of all things, is accepted as a three-faced polyhedron.

This studied prolonging of curtailments serves not merely to prolong an iconoclastic holiday. It affords instructively varied illustrations of the dialectic of conjecture and refutation which Lakatos, following Popper, recognises as the logic of scientific discovery. A mathematical inquiry begins, he says, with a conjecture and, by way of tentative proof, a thought-experiment which decomposes the conjecture into subconjectures. Counter-instances to the original conjecture emerge; the proof is then re-examined in search of a guilty subconjecture, which at length is convicted in turn by a counter-instance. A new, tighter subconjecture is substituted; 'counterexamples are turned into new examples—new fields of inquiry open up'. And, as he well illustrates, the process can repeat itself.

The volume includes three supplements to the original essay, all edited

from Lakatos's Cambridge Ph.D. thesis of 1961. One of these added pieces continues the dialogue on faces, edges, and vertices of polyhedra, concentrating now on Poincaré's proof of Euler's law by methods of vector algebra. It is not easy. A second supplement brings a different case study: no longer of Euler's law, but of a law noted by Leibniz, and professedly proved by Cauchy, to the effect that the limit of a convergent series of continuous functions is a continuous function. The final supplement follows up with further case studies, and at this point the concern is with pedagogical values. Lakatos does not in the end deny the feasibility of full formalistic rigour in mathematical proof, but he makes an eloquent and conclusive case for preferring the heuristic style of conjecture and refutation in mathematical treatises and textbooks.